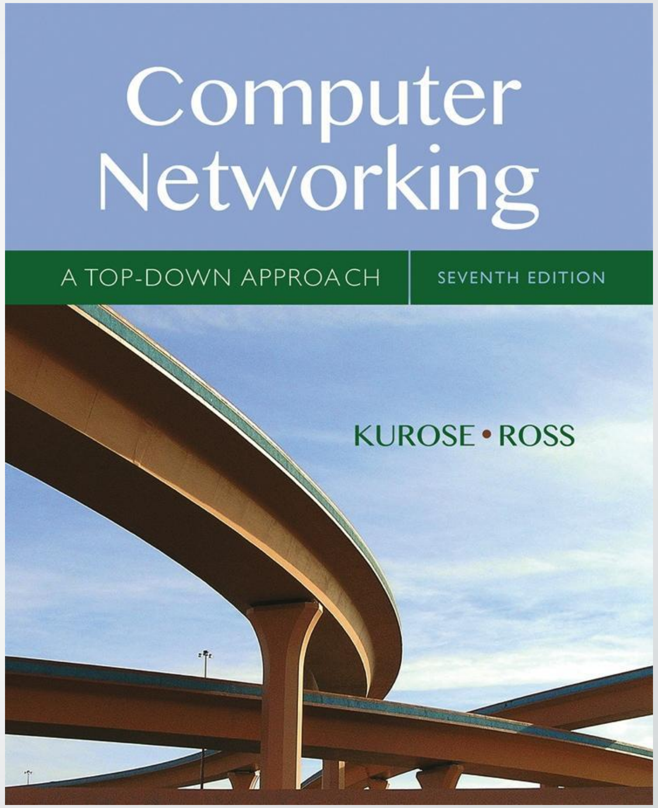
Data Link Layer

Lecture 15 | CSE421 –Computer Networks

Department of Computer Science and Engineering
School of Data & Science

Based on Chapter 6
The Link Layer and LANs

- *The slides are adapted from Kurose and Ross, Computer Networks 7th edition, Kurose and Ross.*

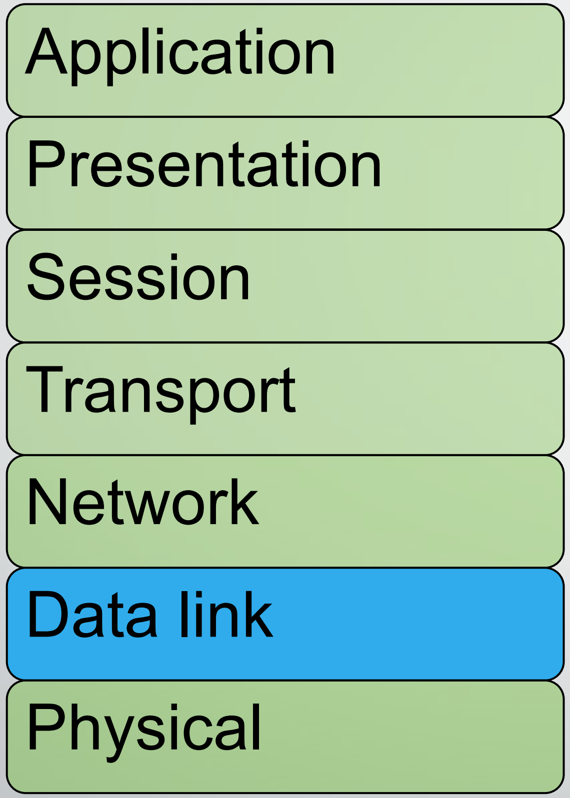


Computer Networking: A Top Down Approach
7^{th} edition
Jim Kurose, Keith Ross
Pearson/Addison Wesley
April 2016

Chapter 6: Link layer and LANs

Objectives:

- understand principles behind link layer services:
  - error detection, correction (done in CSE320)
  - sharing a broadcast channel: multiple access (done in CSE320)
  - Framing - link layer addressing
  - ARP
  - local area networks: Ethernet

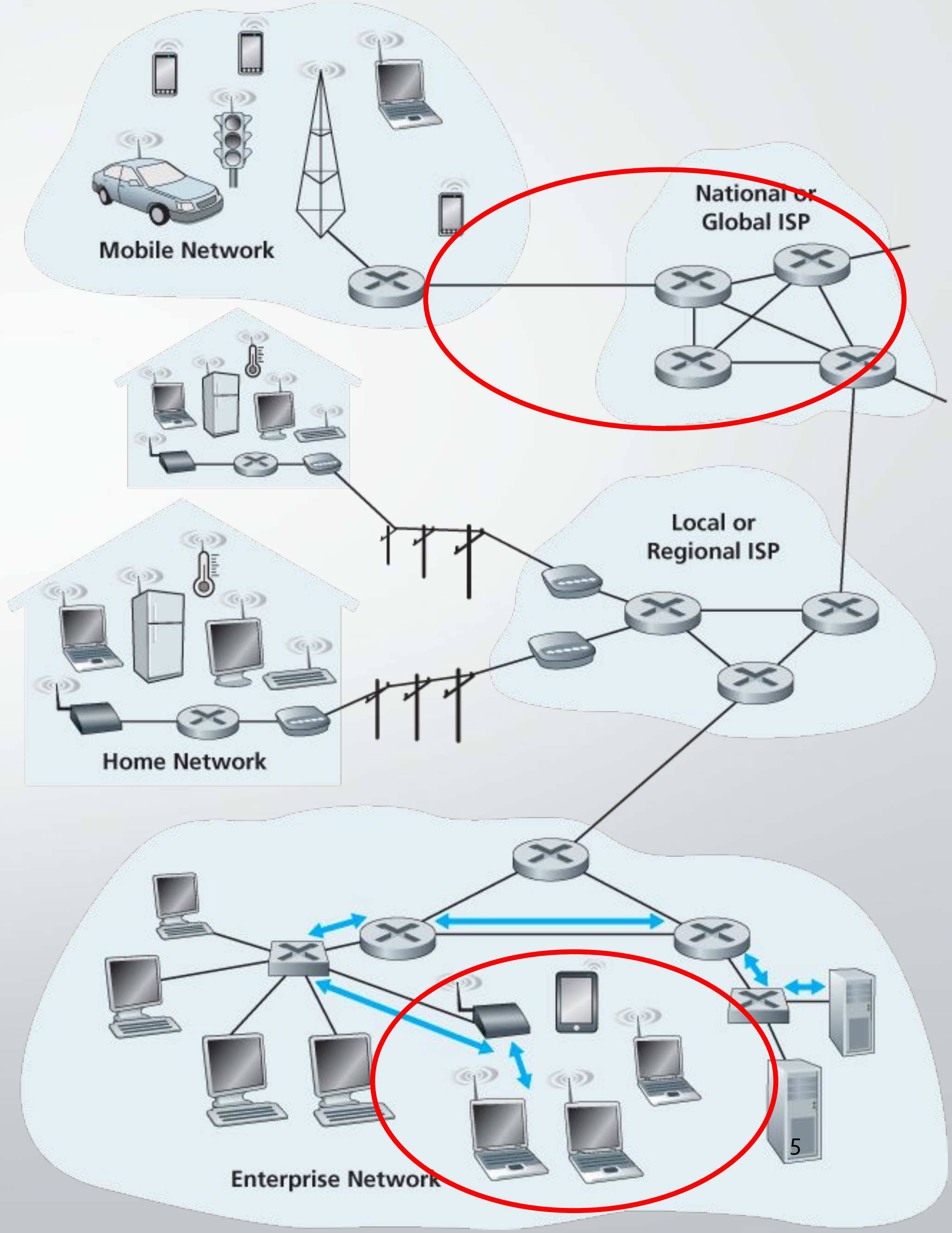


Introduction to Link Layer

Link Layer Terminology

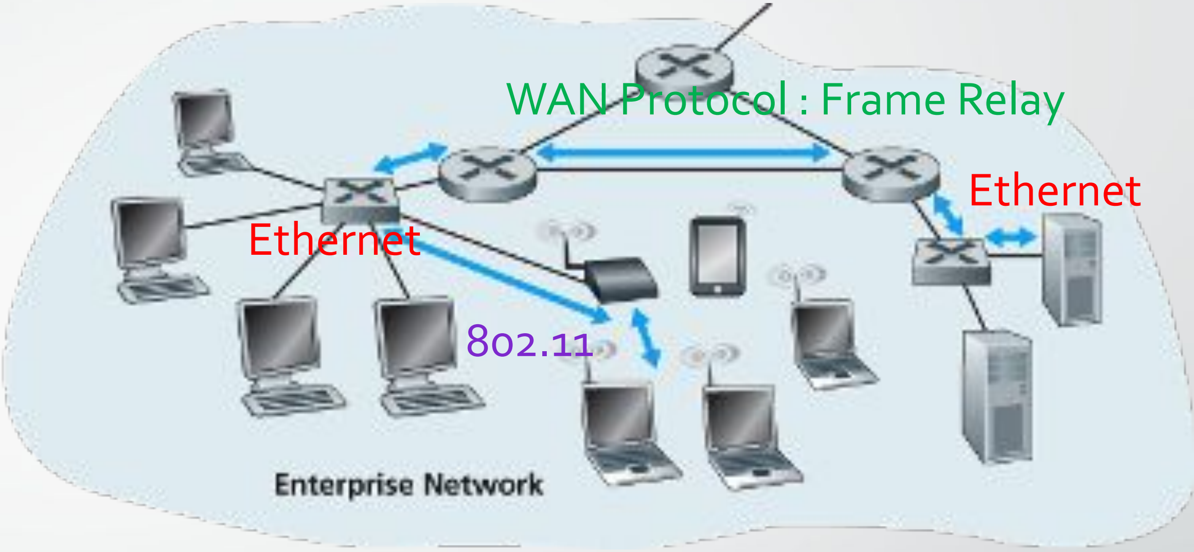
- Nodes : hosts and routers
- Links:
  - wired links
  - wireless links
- Frame : layer-2 packet

The data-link layer is responsible for transferring a datagram from one node to its directly connected (physically adjacent) node over a link.



Link layer: context

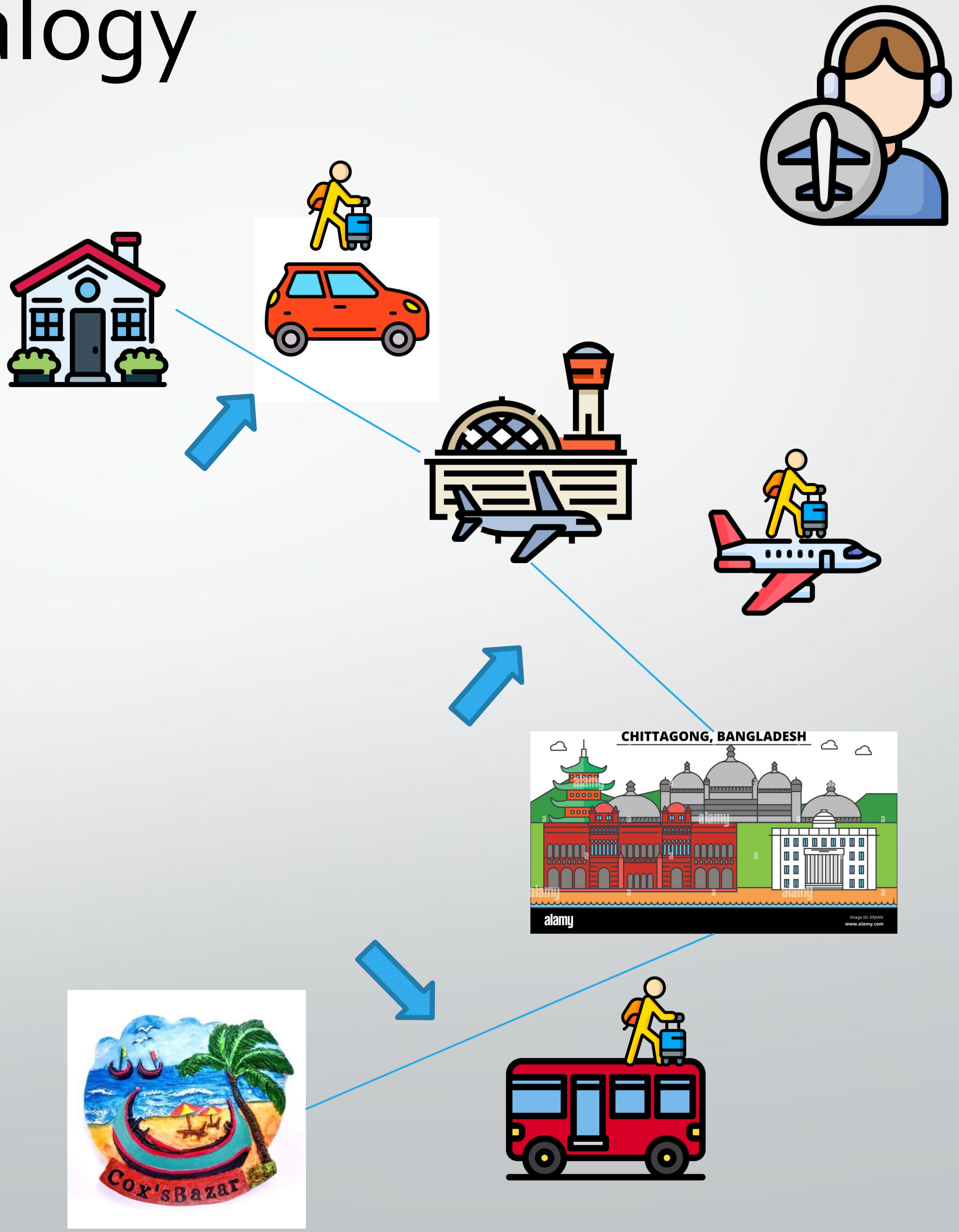
- As data moves through a network, it can travel over many types of links.
  - e.g., Ethernet on first link, frame relay on intermediate links, 802.11 on last link
- Each link uses its own rules (protocols) to send the data.



Analogy

Transportation analogy:

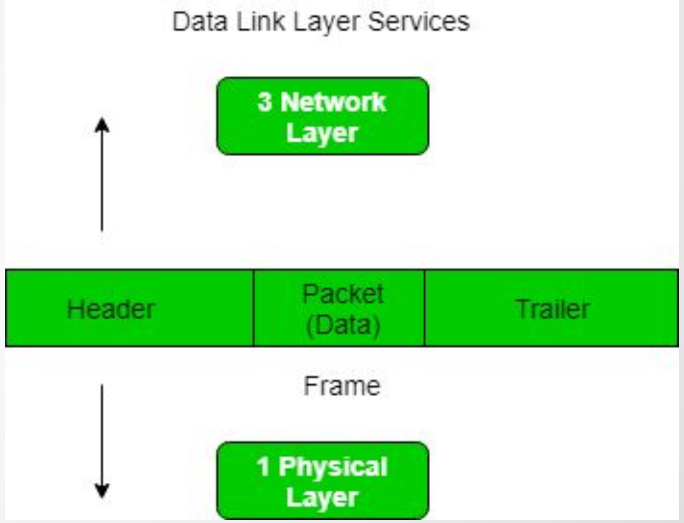
- trip from Home to Cox’s Bazaar
  - Uber Car : Home to Dhaka Airport
  - Plane: Dhaka to Chittagong
  - Bus: Chittagong to Cox’s Bazaar
- tourist = datagram
- transport segment = communication link
- transportation mode = link layer protocol
- travel agent = routing algorithm

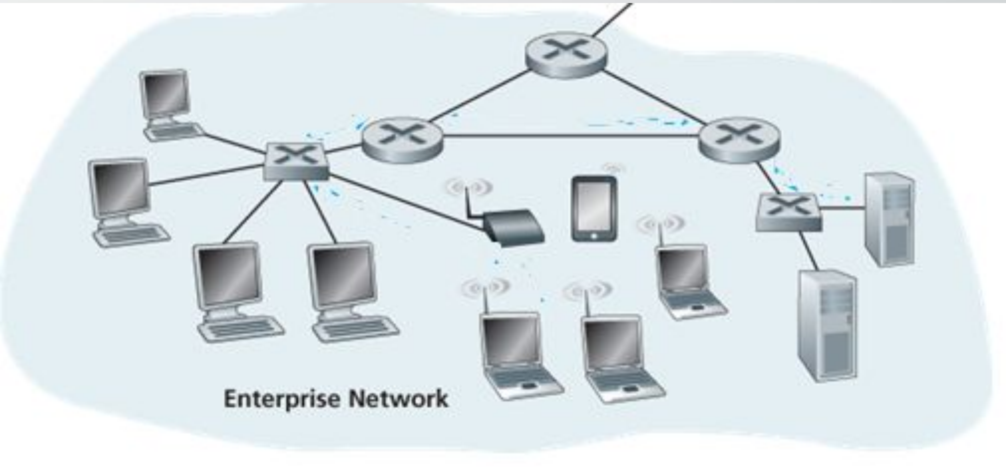


Link layer functions/services

- *Framing*
  - Breaks data into frames (with header + trailer).
  - "MAC" addresses used in frame headers to identify source, destination devices.

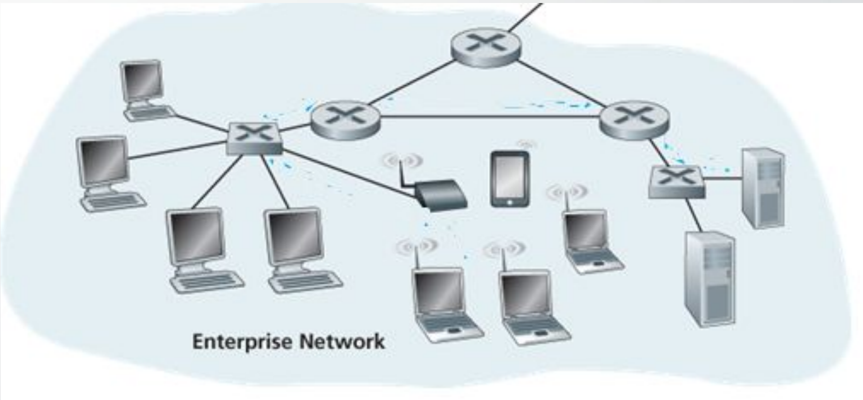
- *Link access:*
  - Decides how to send a frame to the link
  - Manages sharing if many devices use the same channel.
  - Prevents or controls collisions in multi-user networks.
  - Sets rules for sending data on the link.





Link layer functions/services

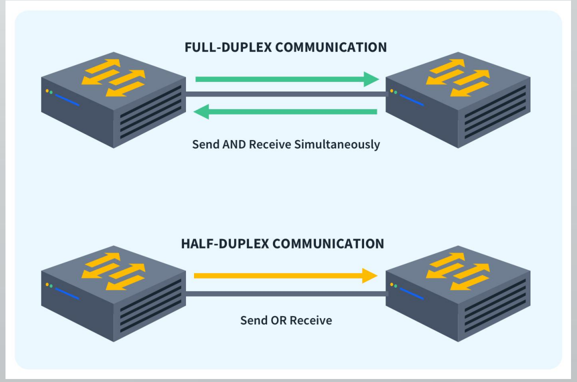
- *Reliable delivery between adjacent nodes*
  - Makes sure frames are delivered correctly to the next device.
  - Seldom used on low bit-error link (fiber, some twisted pair)
  - Important on wireless links: high error rates
    - *Q:* why both link-level and end-end reliability?



- *Flow control*
  - Manages the **speed of data** between sender and receiver.
  - Prevents the sender from overwhelming the receiver.

Link layer services (more)

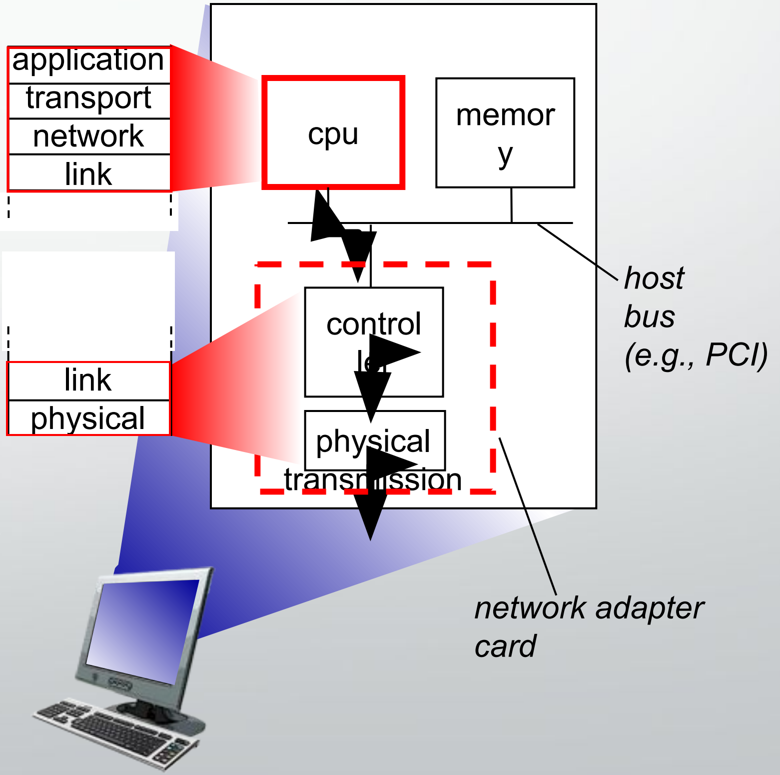
- *Error detection:*
  - errors caused by signal attenuation, noise.
  - receiver detects presence of errors:
    - signals sender for retransmission or drops frame
- *Error correction:*
  - receiver identifies *and corrects* bit error(s) without resorting to retransmission (there are various protocols)
- *Full Duplex and Half Duplex:*
  - Both devices can send and receive **at the same time**
  - Both devices can send and receive, but **only one at a time**



Where is the link layer implemented?

- Link layer runs in the adaptor (NIC) or chipset
  - Ethernet card
  - 802.11 (WiFi card)
- Provides both link-layer and physical-layer functions
- Connects the computer to the network (via system bus)
- Combination of hardware, software, firmware

Every device has one



Our objectives

Objectives – Part 1

- Link Layer Addressing
  - MAC Address
  - Types of MAC Addresses
- ARP
- ARP within LAN
- LAN Switch

Link Layer Addressing

MAC Address

- A **MAC address (Media Access Control address)** is a unique identifier assigned to a network interface card (NIC) or adaptor.
- **Layer:** Works at the **Data-Link Layer** (Layer 2 of the OSI model).
- **Purpose:** Used for hop to hop communication

- **Uniqueness:** Each device's NIC has a unique MAC address worldwide.
- **Format:** 48 bits (6 bytes), usually written as 6 pairs of hex numbers separated by colons or hyphens.
  - Example : 1A:2F:BB:76:09:AD

IP Address vs MAC Address

IP address

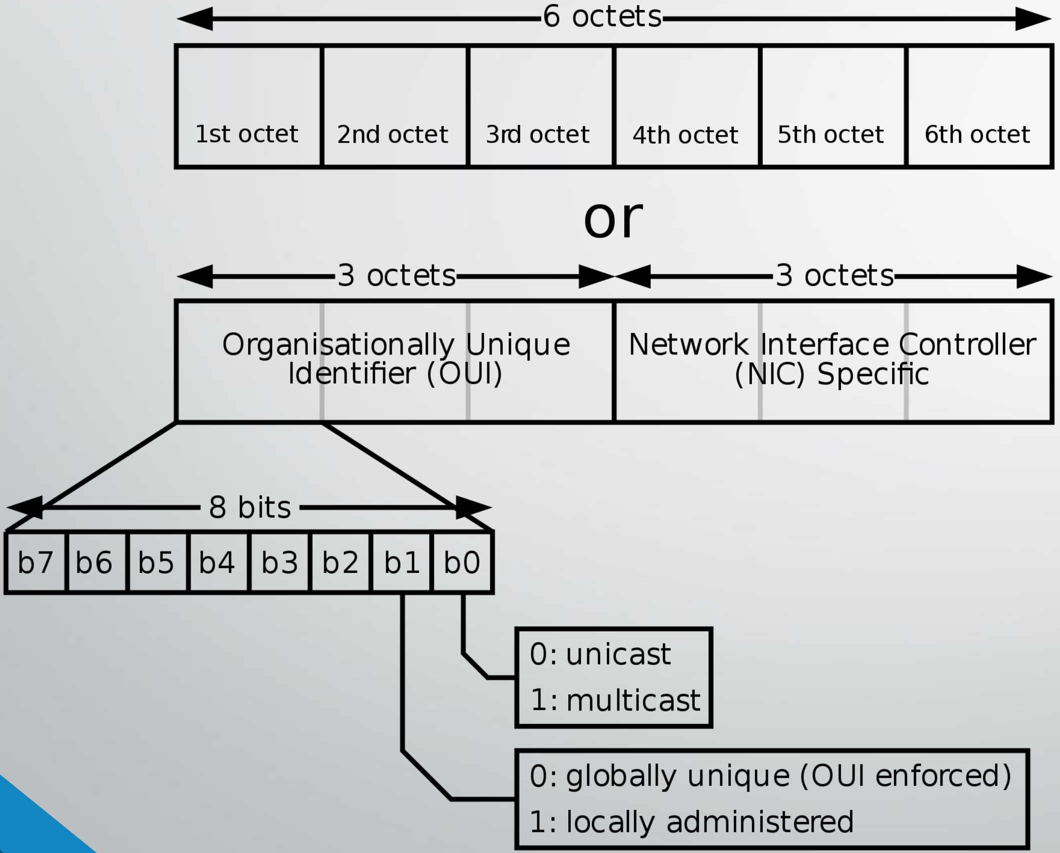
- 32 bits
- Dotted decimal notation
  - Example : 192.168.10.1
- *Network-layer* address for interface
- Hierarchical
  - Not portable
- Function

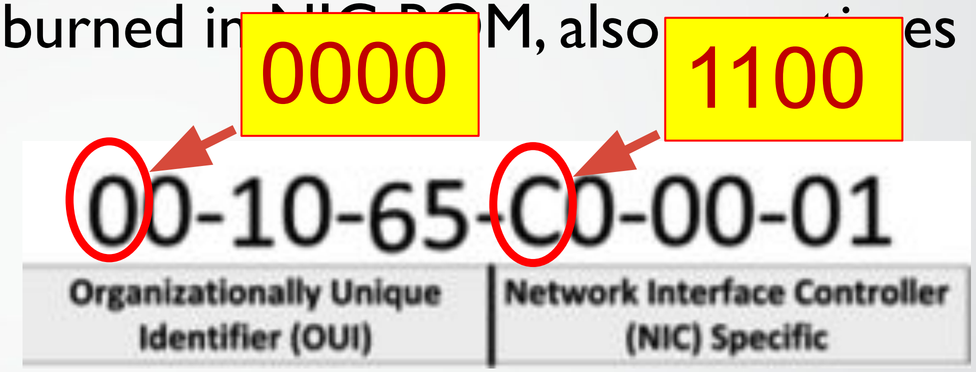
MAC address

- 48 bits
- 12 Hexadecimal digits
  - Example : 1A-2F-BB-76-09-AD
- *Data Link-layer* address for interface
- Flat
  - portable
- Function

MAC Address

- 48 bits MAC address (for most LANs) burned in [...]M, also [...]es software settable



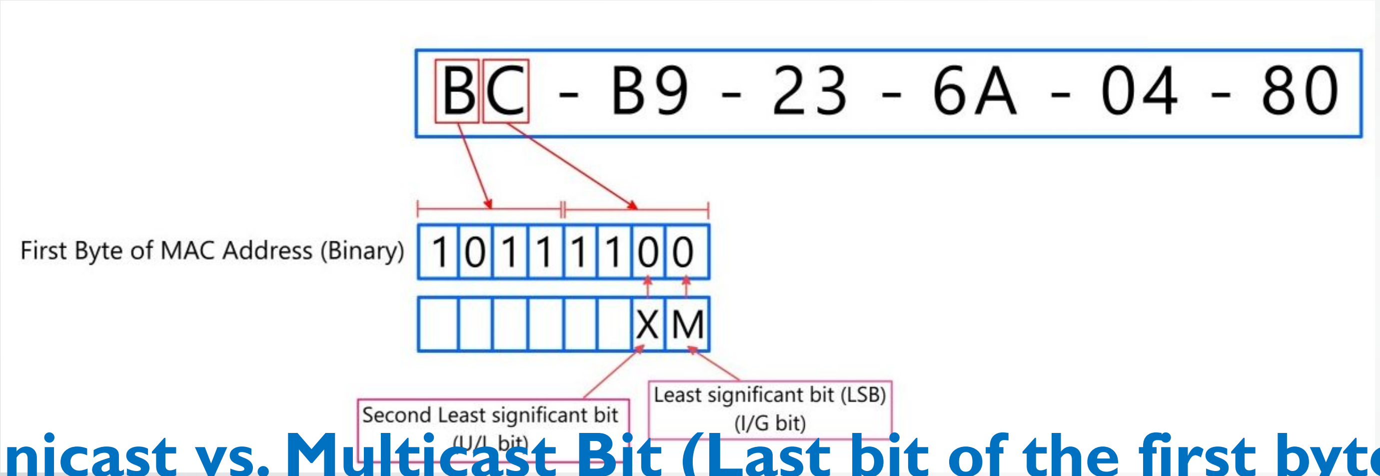


hexadecimal (base 16) notation
(each "numeral" represents 4 bits)

Different display formats:

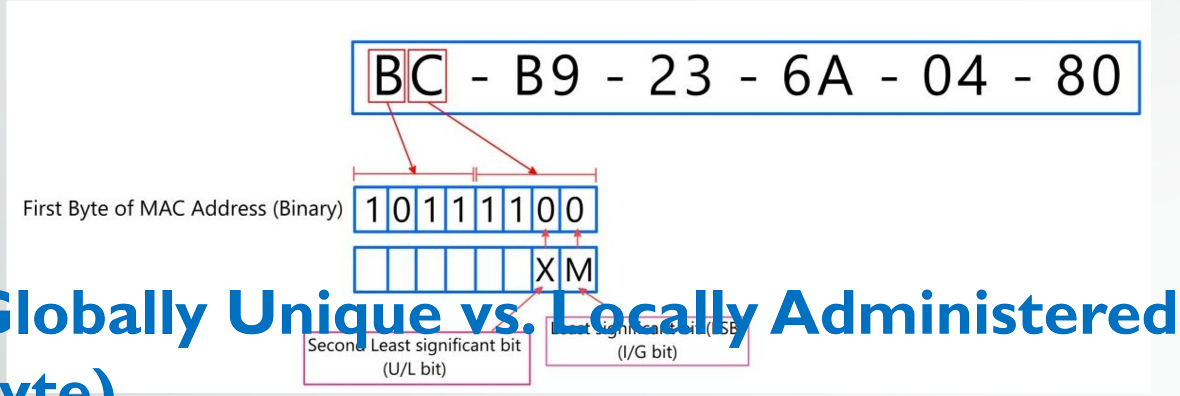
- 0000.0c43.2e08
- 00:00:0c:43:2e:08
- 00-00-0C-43-2E-08

MAC Address



- **Unicast vs. Multicast Bit (Last bit of the first byte)**
  - This bit decides **who the MAC address is meant for**.
- If the **last bit of the first byte = 0,** the MAC address is **unicast**.
  - Means: the address belongs to one single device.
- If the **last bit of the first byte = 1**, the MAC address is **multicast.**
  - Means: the frame is delivered to a group of devices.

MAC Address



- **Globally Unique vs. Locally Administered (Second-last bit of the first byte)**
  - This bit decides **who gave the address**.
- If **the bit = 0**, the MAC address is **assigned by the manufacturer**.
  - The MAC address is guaranteed to be **Globally Unique**.
- If **the bit = 1**, the MAC address is **changed manually** (by an admin or software).
  - Used when you want to override the default.

MAC or LAN or Physical or Ethernet addresses (more)

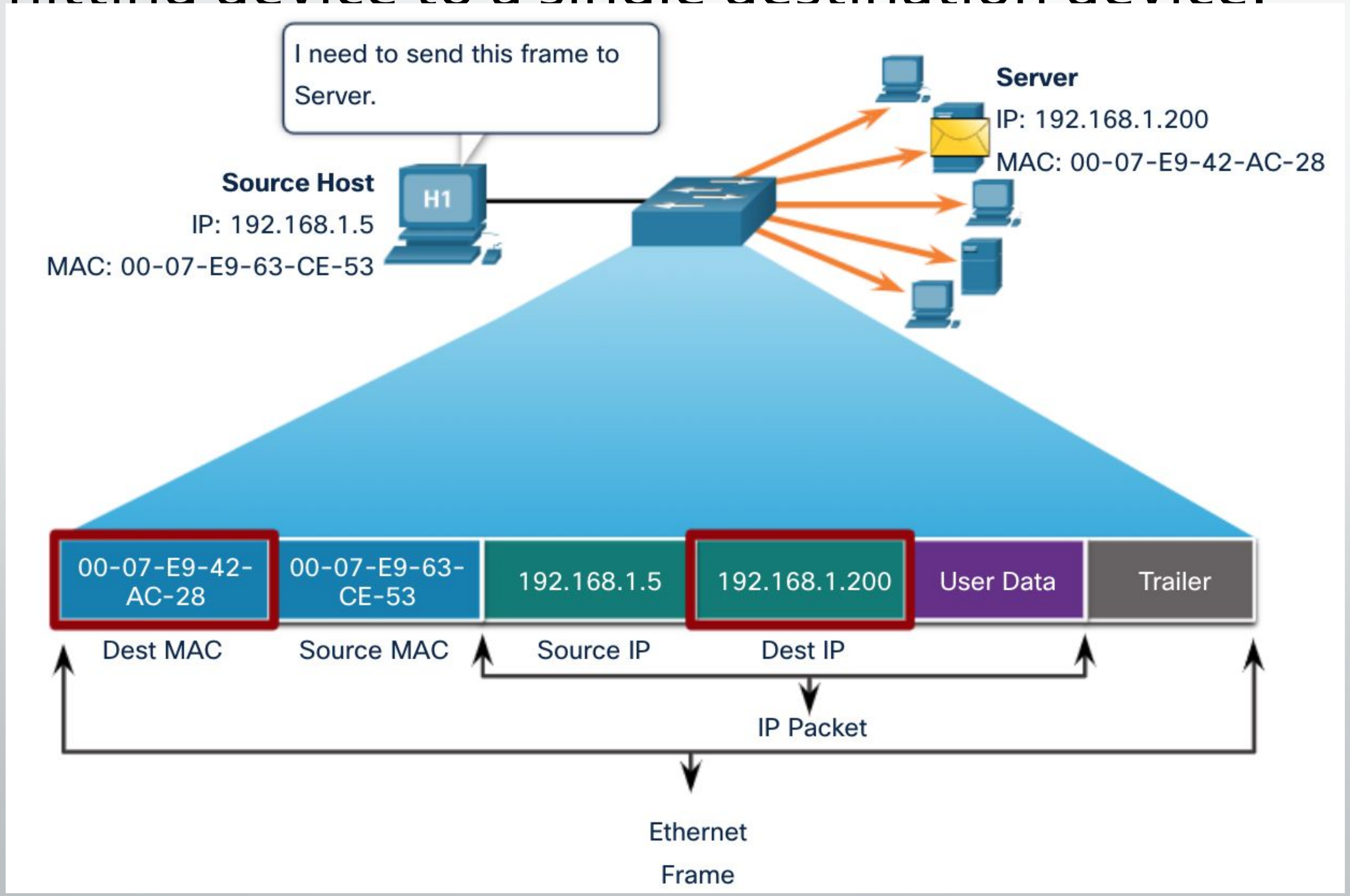
- 48 bits MAC address (for most LANs) burned in NIC ROM, also sometimes software settable
- MAC address allocation administered by IEEE
- Manufacturer buys portion of MAC address space (to assure uniqueness)
- Analogy:
  - MAC address: like National ID
  - IP address: like Postal Address

Types of MAC Address

- MAC Address
  - Unicast
  - Multicast
  - Broadcast

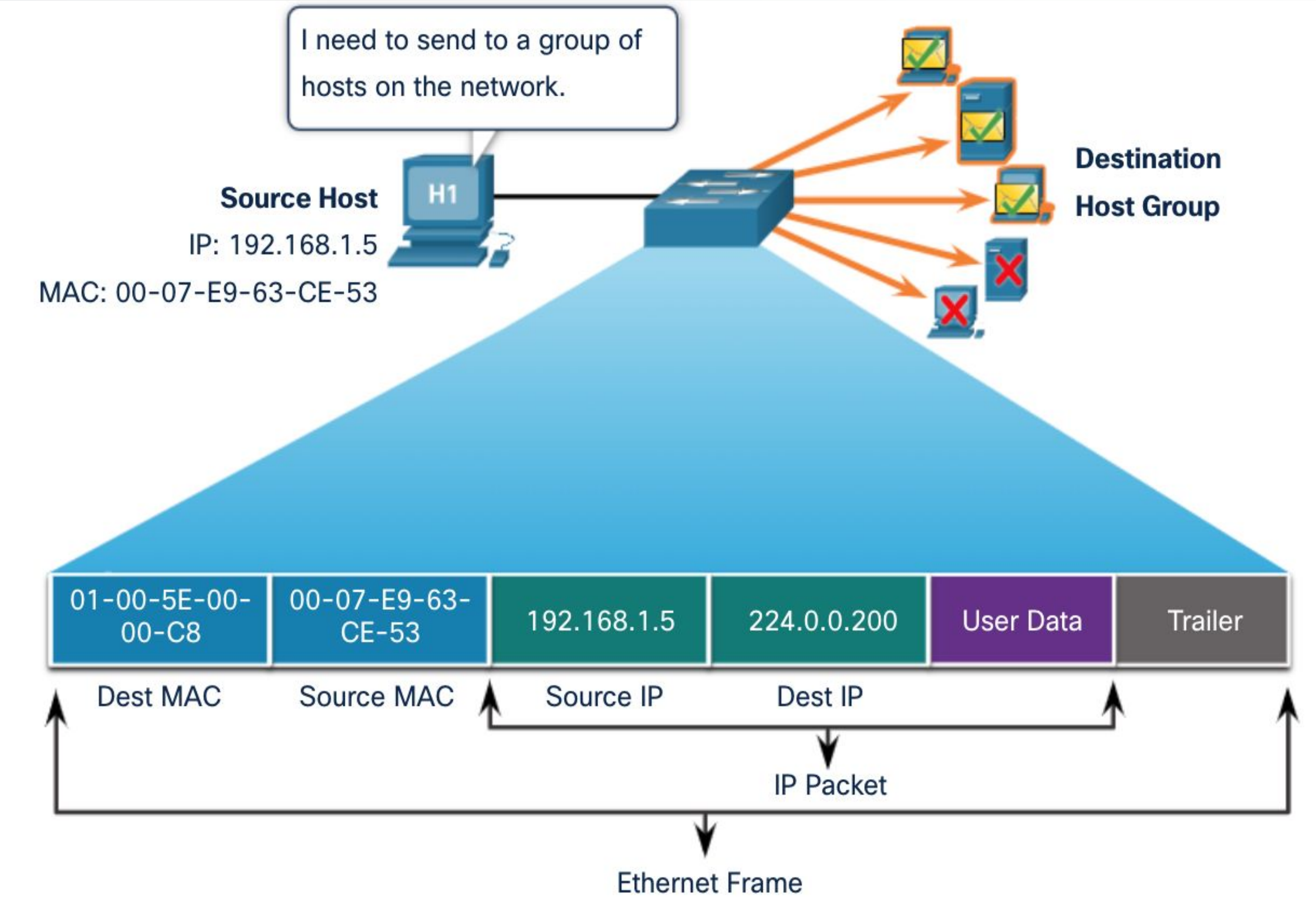
Unicast MAC Addresses

- The unique address used when a frame is sent from a single transmitting device to a single destination device.



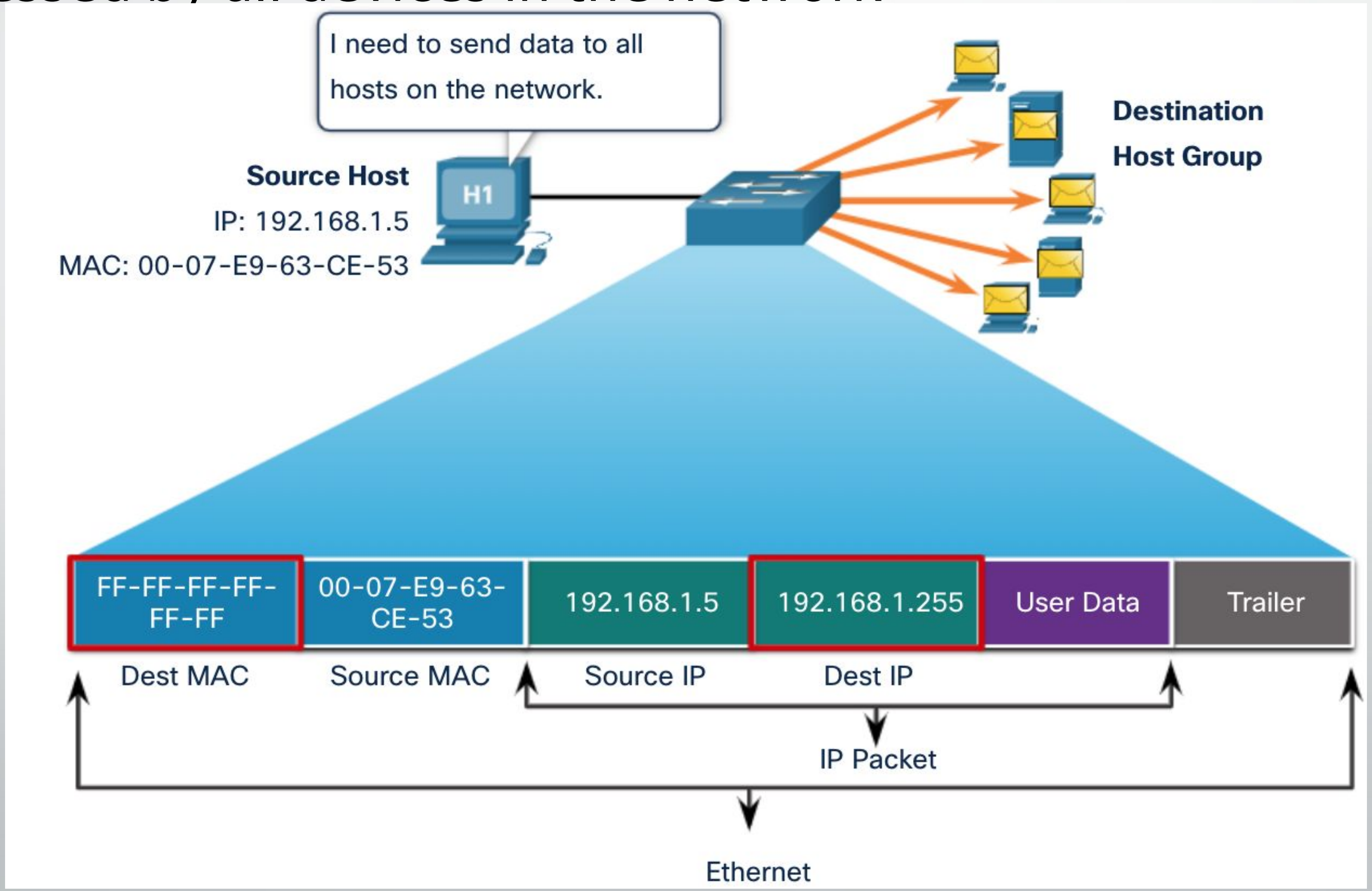
Multicast MAC Addresses

- " 01-00-5E" in an IPv4 multicast packet



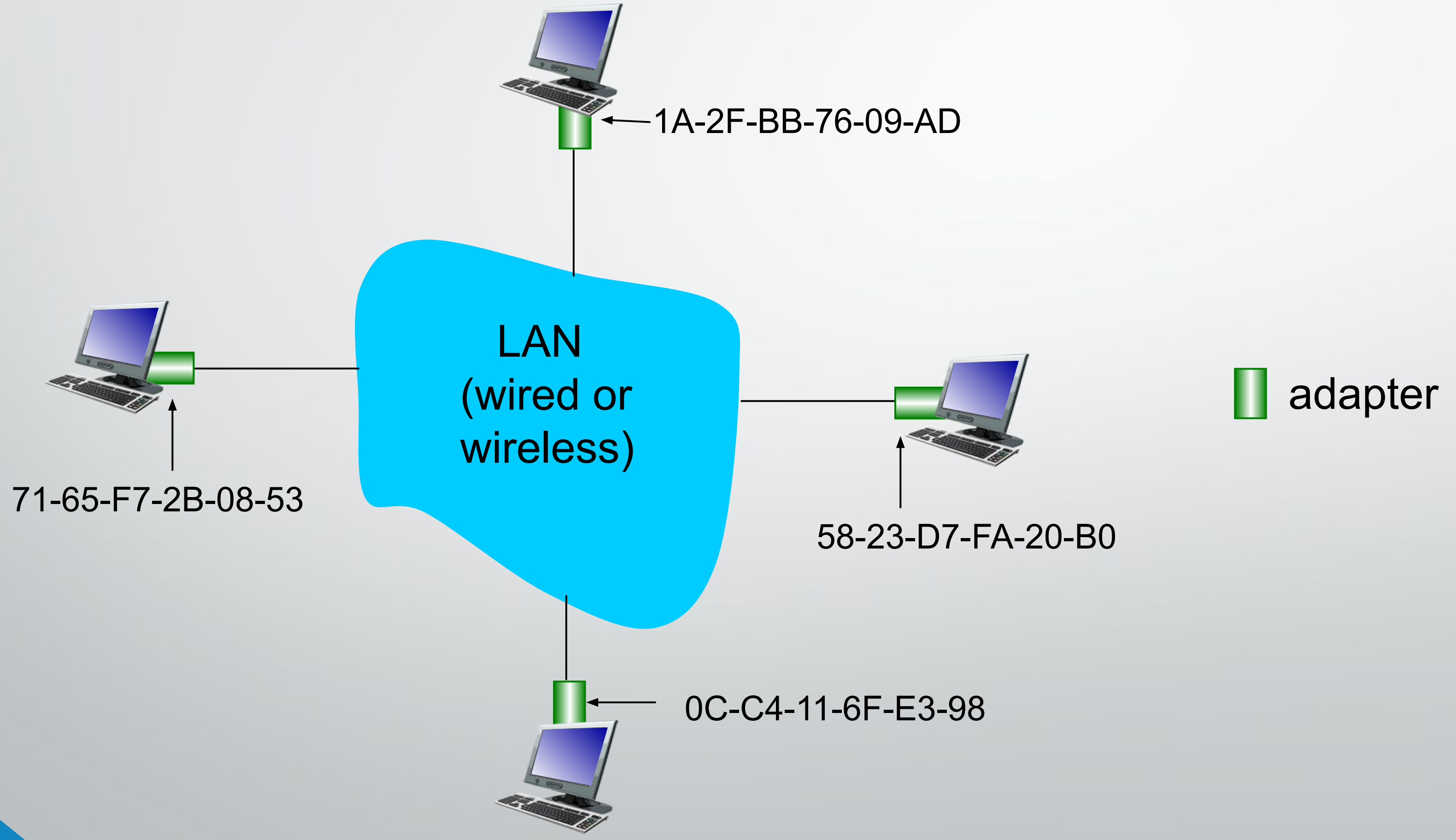
Broadcast MAC Address

- A destination MAC address of FF-FF-FF-FF-FF-FF
- To be processed by all devices in the network



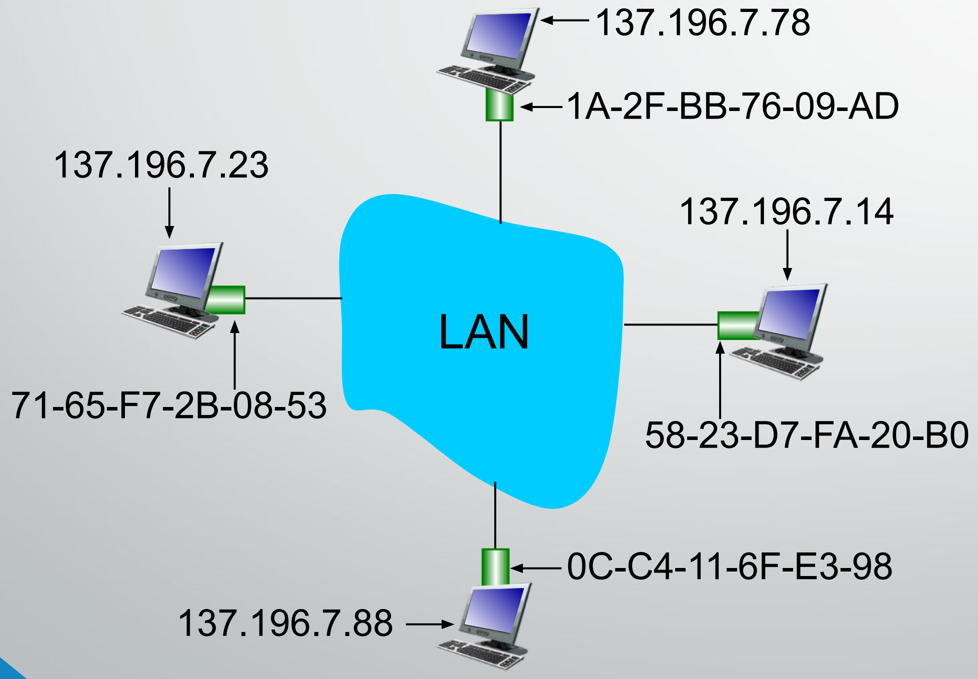
LAN addresses and ARP

each adapter on LAN has unique *LAN* address



ARP: address resolution protocol

Question: how to determine interface's MAC address, knowing its IP address?



- *ARP*
- *Mapping _IP Add___ to _MAC Add____*
- *ARP table*
  - IP
  - MAC address
  - TTL (Time To Live) Or Age
    - time after which address mapping will be forgotten (typically 20 min)

ARP Tables

```
C:\WINDOWS\system32\cmd.exe

C:\>arp -a

Interface: 192.168.0.2 --- 0x2
  Internet Address      Physical Address      Type
  192.168.0.1           00-0a-cd-00-0d-1d     dynamic

C:\>
```

IP Adress

MAC Adress

ARP Type

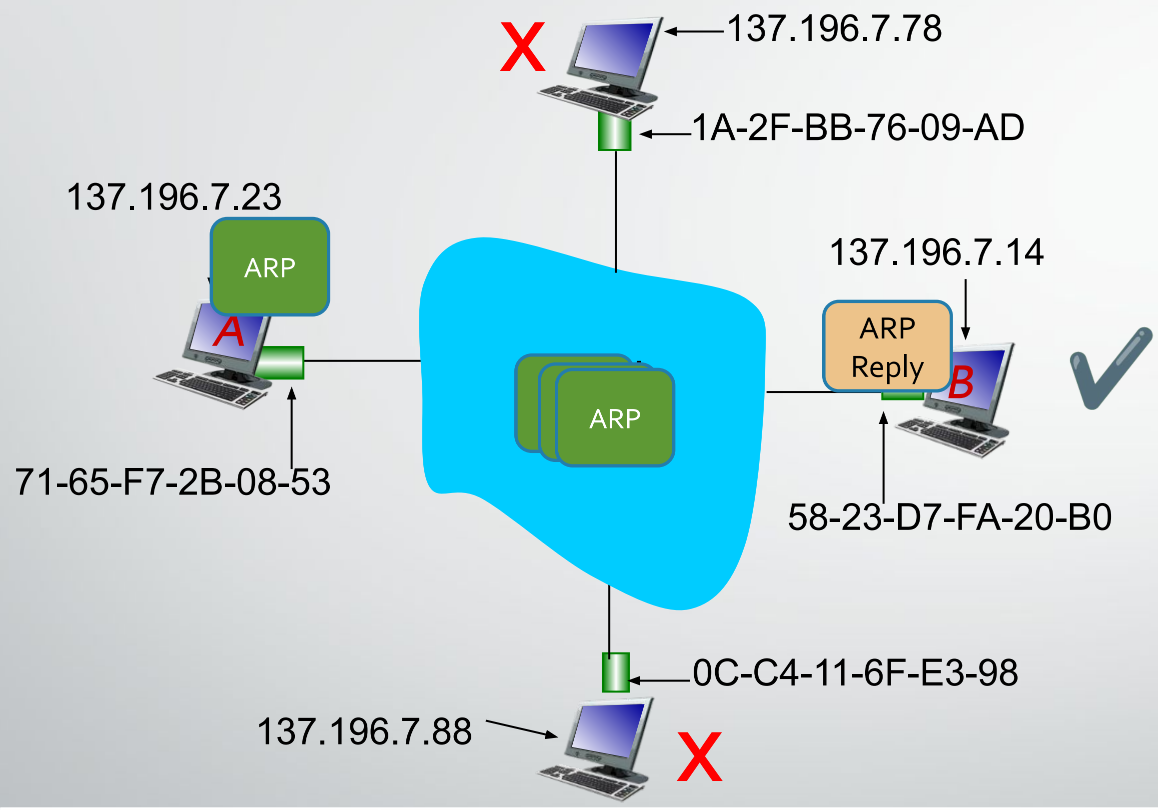
Host or PC

R1#sh ip arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.1	-	ca02.238f.0008	ARPA	FastEthernet0/0
Internet	192.168.1.2	6	0050.7966.6800	ARPA	FastEthernet0/0
Internet	192.168.2.1	-	ca02.238f.0006	ARPA	FastEthernet0/1
Internet	192.168.2.2	6	0050.7966.6801	ARPA	FastEthernet0/1

Router

ARP: address resolution protocol



A wants to send datagram to B

71-65-F7-2B-08-53	58-23-D7-FA-20-B0	137.196.7.23	137.196.7.14	ARP Reply
FF-FF-FF-FF-FF-FF	71-65-F7-2B-08-53	137.196.7.14	137.196.7.23	ARP Request
Dest MAC	Source MAC	Dest IP Add	Source IP Add	ARP Req/Reply

ARP protocol: same LAN

- A wants to send datagram to B
  - B's MAC address not in A's ARP table.
- A broadcasts ARP query packet, containing B's IP address
  - destination MAC address = FF-FF-FF-FF-FF-FF
  - all nodes on LAN receive ARP query
- B receives ARP packet, replies to A with its (B's) MAC address
  - frame sent to A's MAC address (unicast)

- A caches (saves) IP-to-MAC address pair in its ARP table
  - Soft state: information that times out (goes away) unless refreshed
- ARP is "plug-and-play":
  - nodes create their ARP tables *without intervention from net administrator*

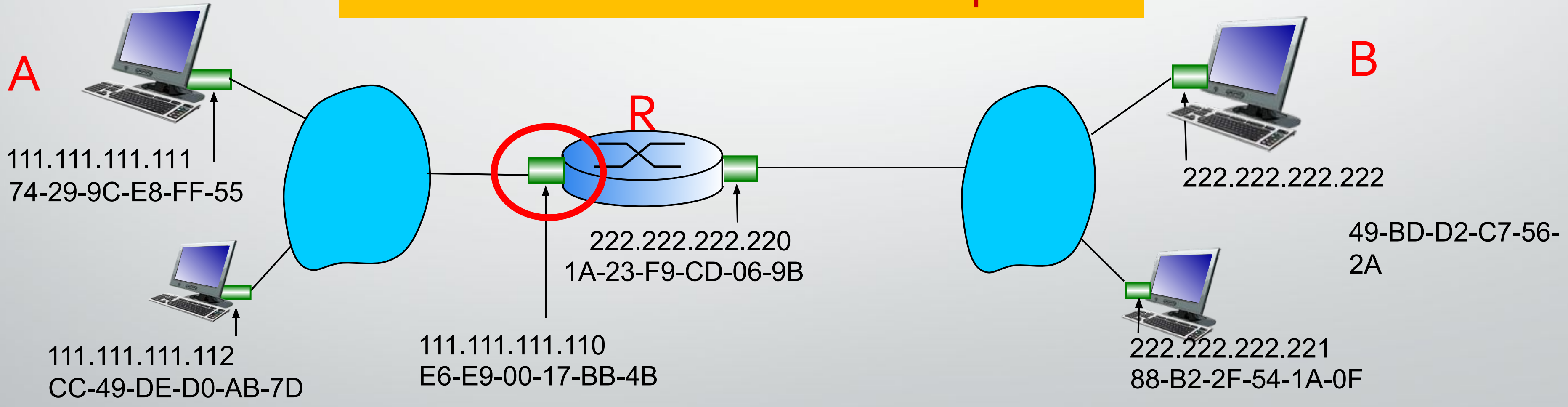
Addressing: routing to another LAN

Send datagram from A to B via R

- focus on addressing – at IP (datagram) and MAC layer (frame)
- assume A knows B's IP address
- What will be the destination MAC Address?

ARP- To know B's MAC address as B's IP address is known

Routers never forward broadcast packets!

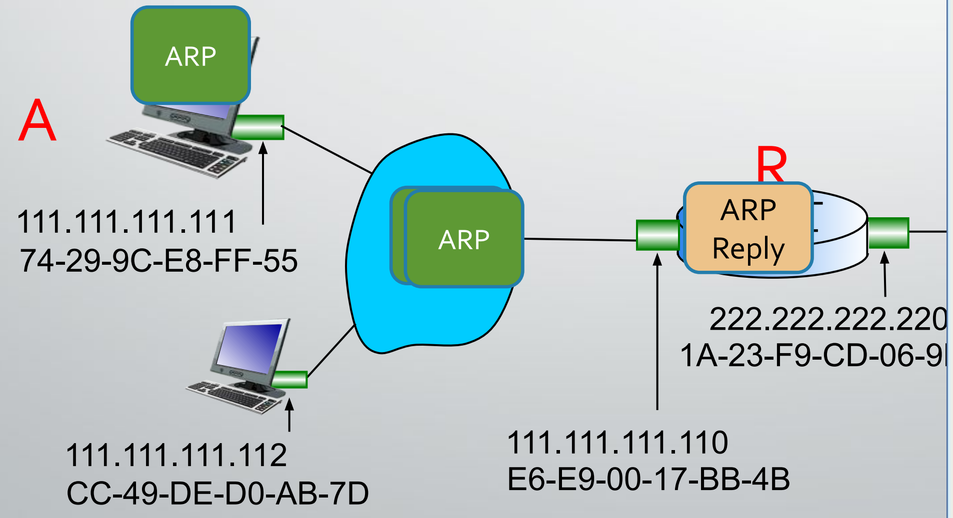


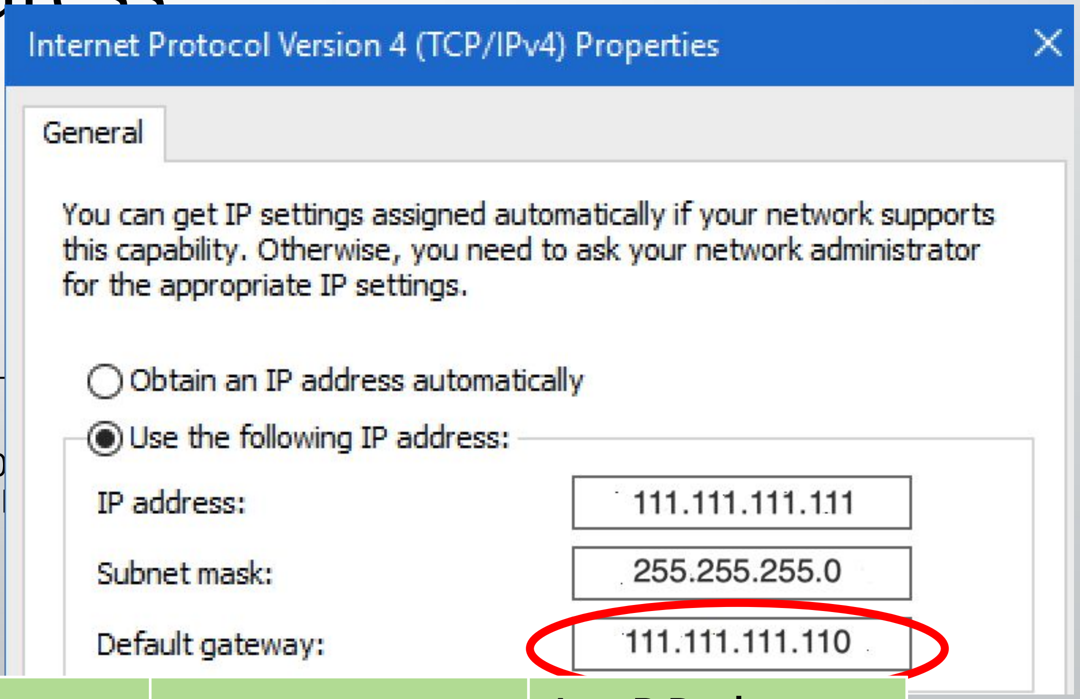
??? Default Gateway's MAC Address	74-29-9C-E8-FF-55	222.222.222.222	111.111.111.111	A to B Packet
Dest MAC	Source MAC	Dest IP Add	Source IP Add	Packet Type

Addressing: routing to another LAN

Send datagram from A to B via R

- Does A know the IP address of first hop router, R which is also known as **Default Gateway**? (how?)
- Will A know R's MAC address?

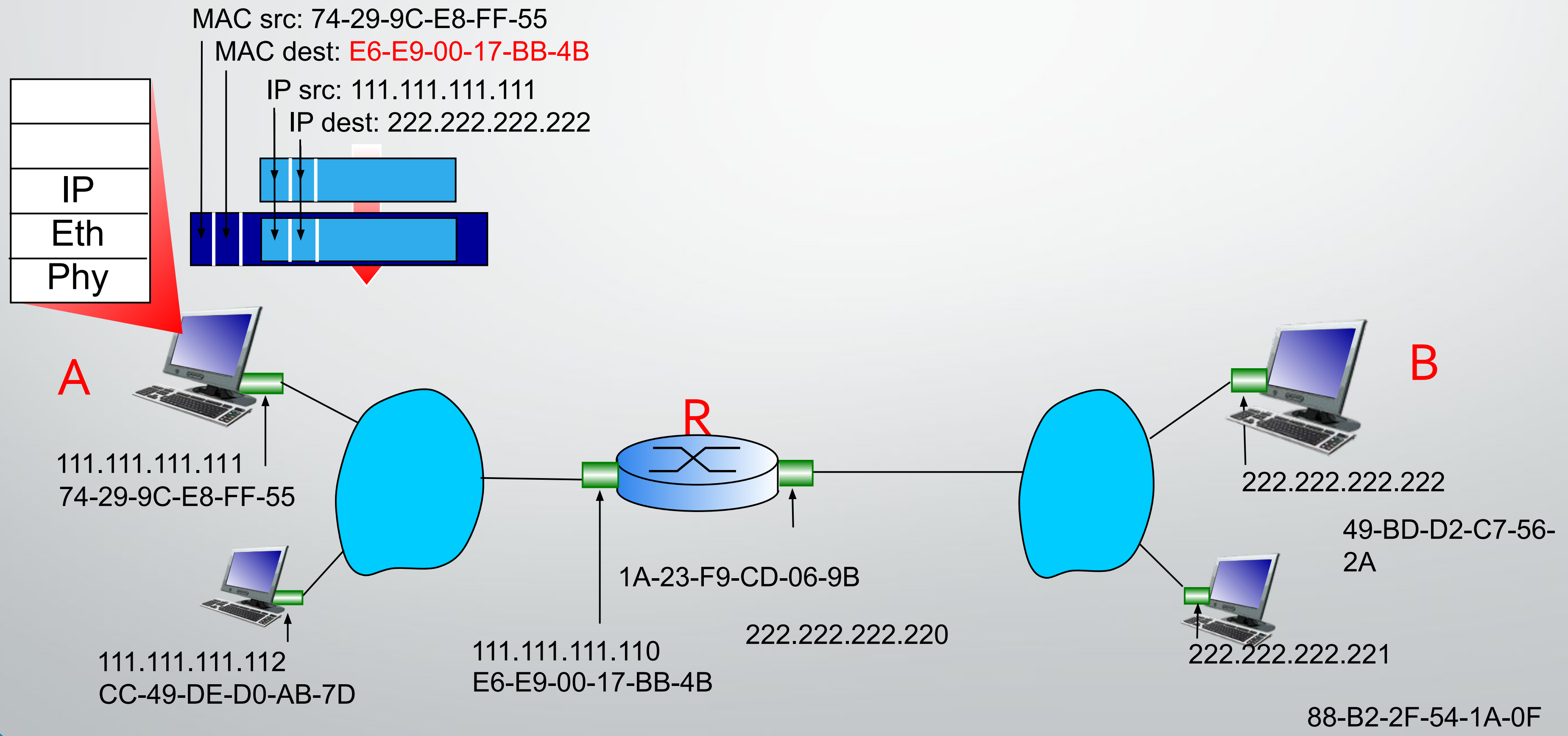




E6-E9-00-17-BB-4B	74-29-9C-E8-FF-55	222.222.222.222	111.111.111.111	A to B Packet

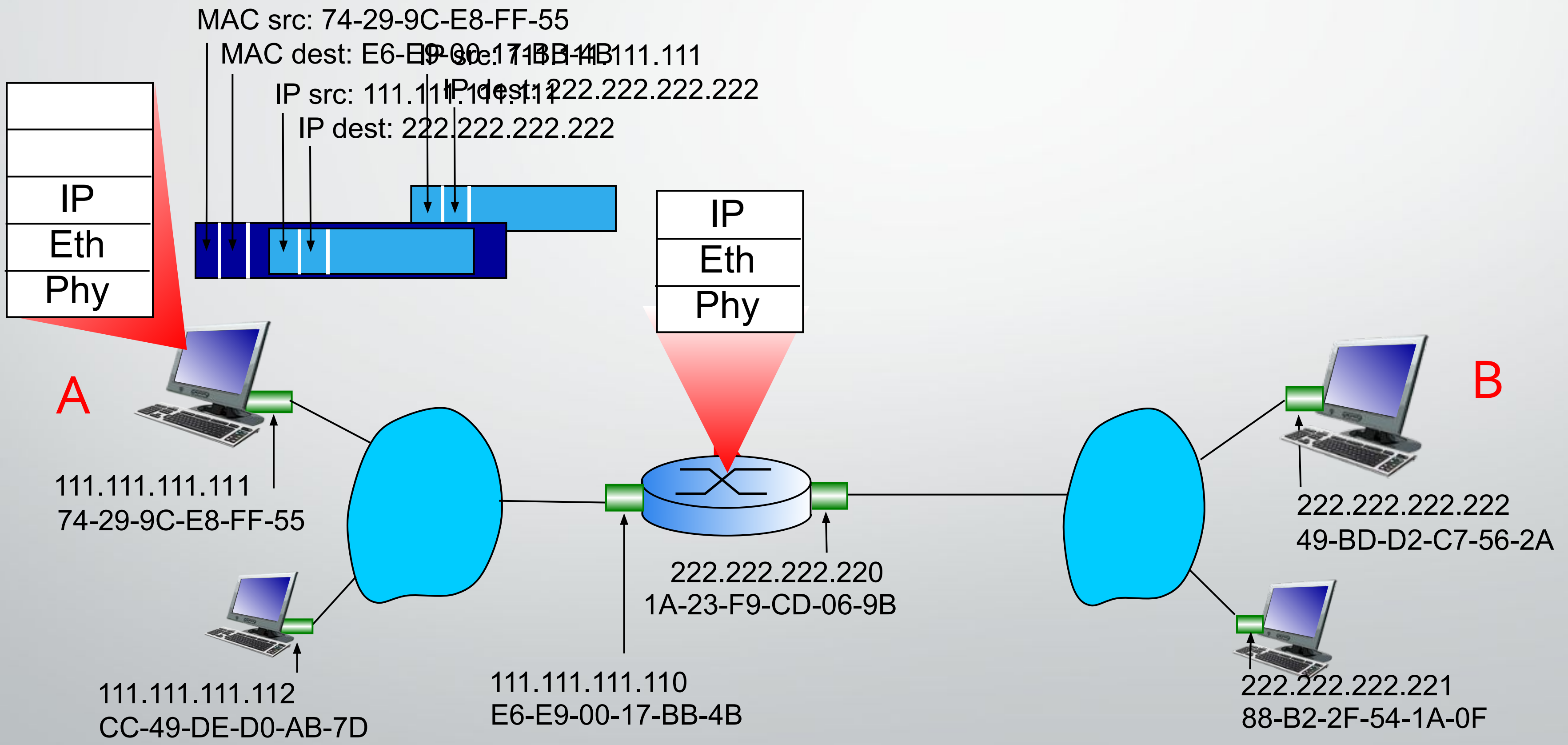
Addressing: routing to another LAN

- A creates IP datagram with IP source A, destination B
- A creates link-layer frame with R's MAC address as destination address, frame contains A-to-B IP datagram



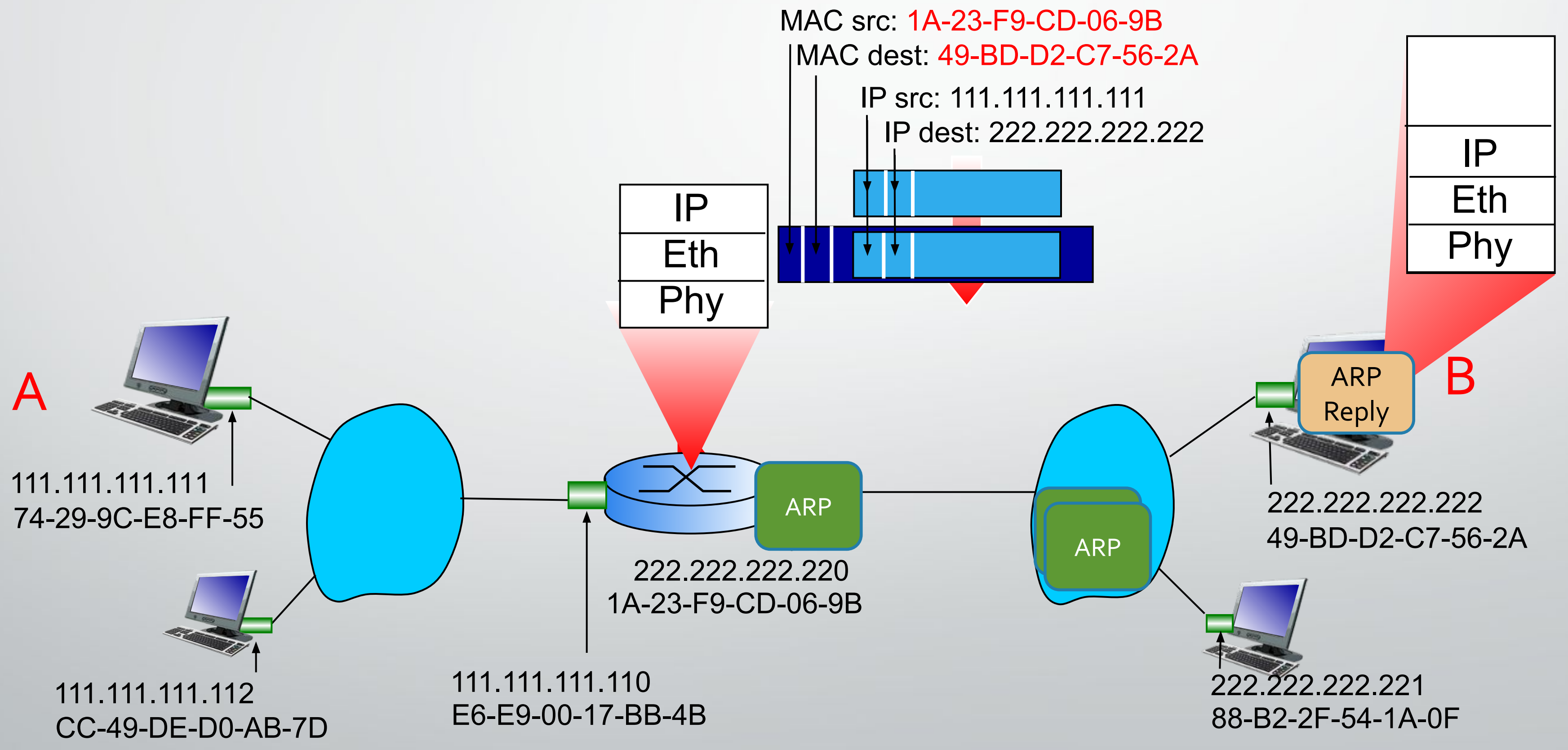
Addressing: routing to another LAN

- frame sent from A to R
- frame received at R, datagram removed, passed up to IP



Addressing: routing to another LAN

- R forwards datagram with IP source A, destination B
- R creates link-layer frame with B's MAC address as destination address, frame contains A-to-B IP datagram



Addressing: routing to another LAN

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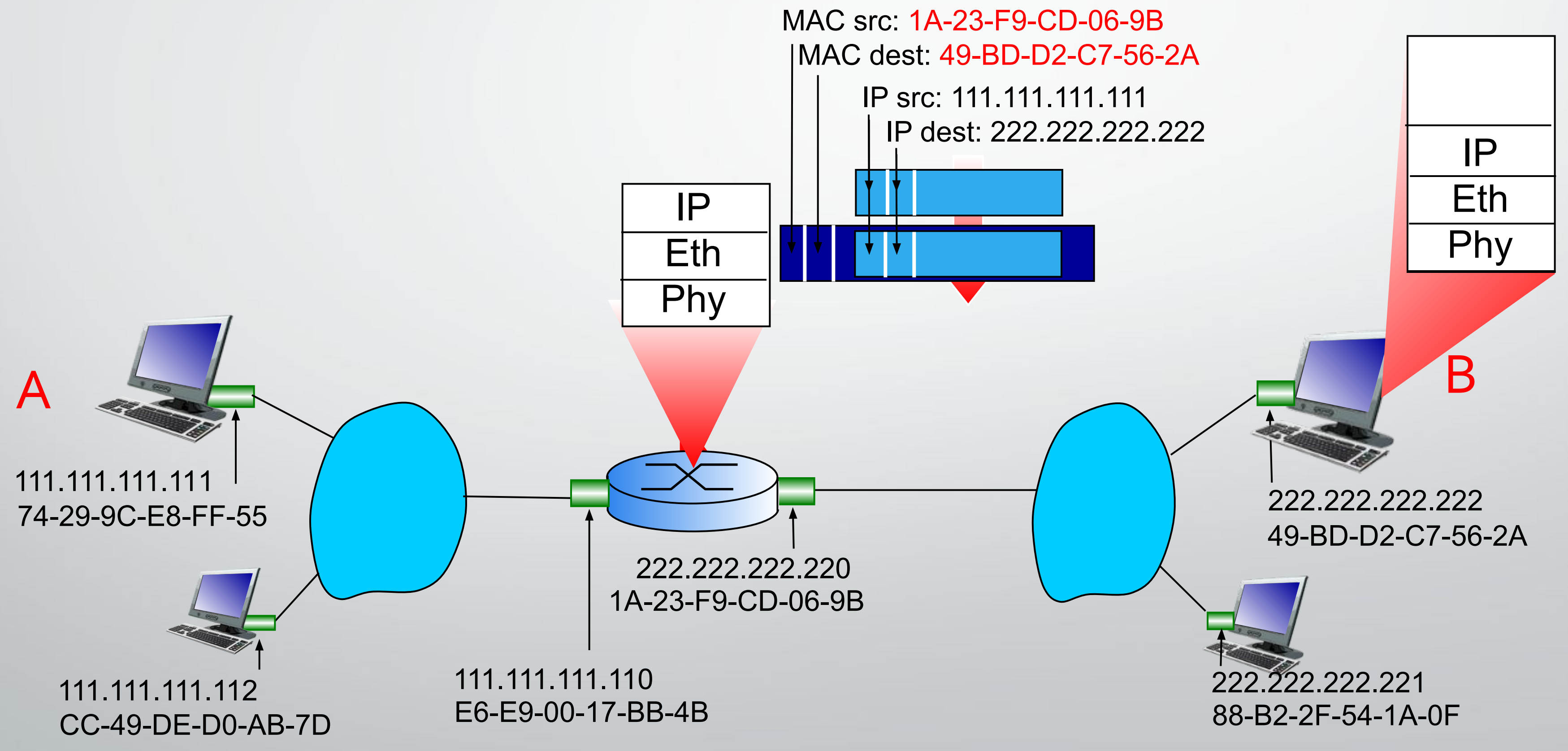

Addressing: routing to another LAN

- R forwards datagram with IP source A, destination B
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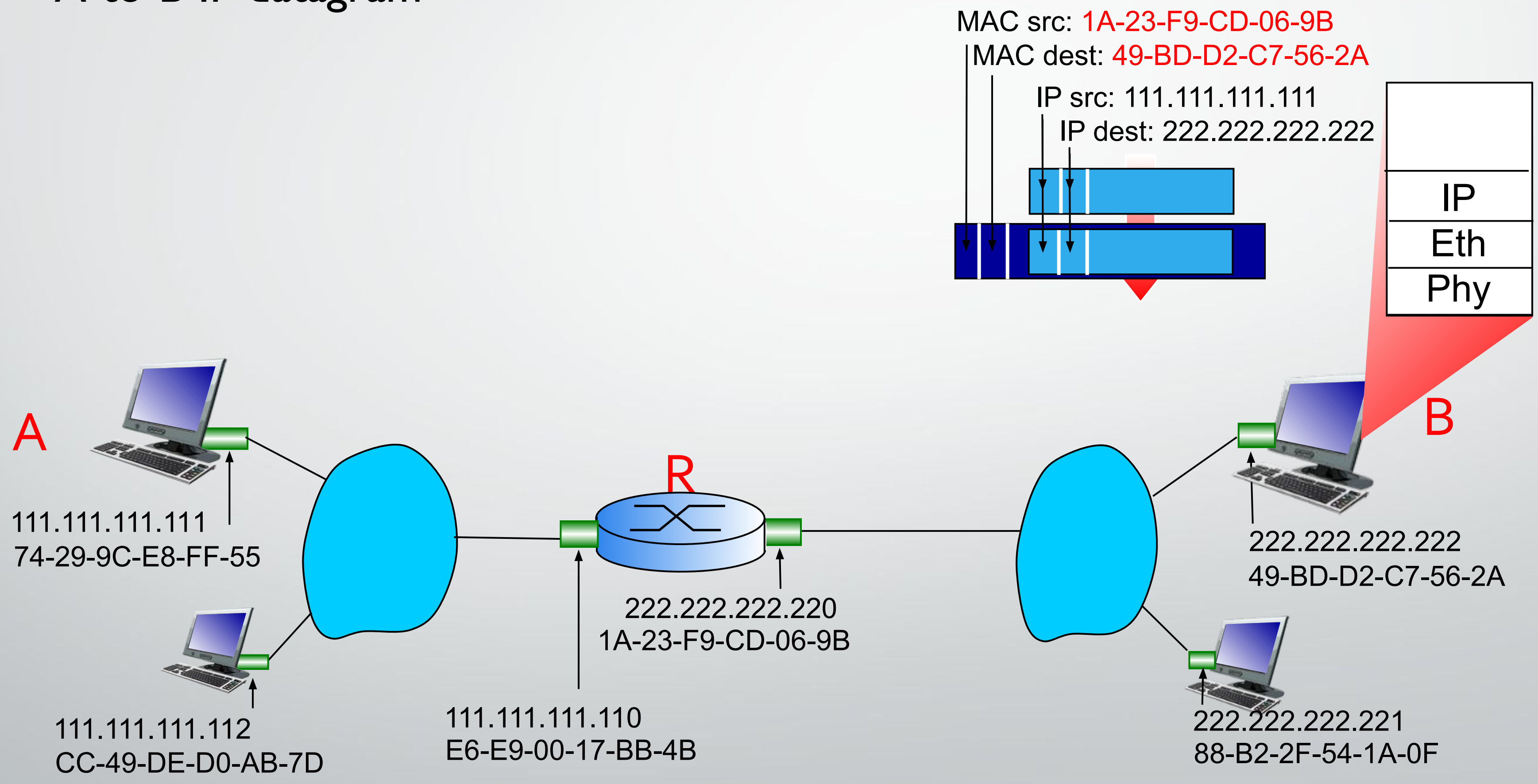


* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

Objectives

- Switch
  - Characteristics of a switch
  - Role of switch in a LAN

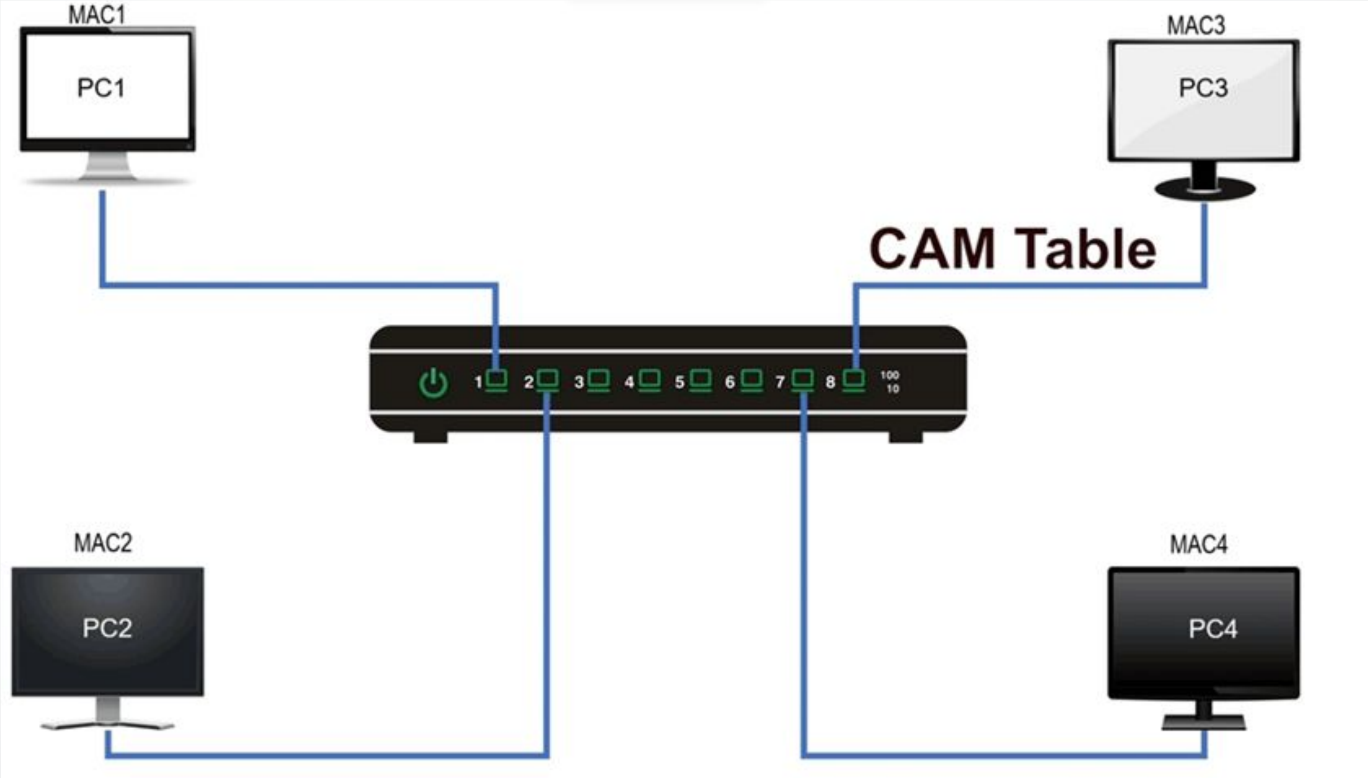
Switch

- link-layer device: takes an *active* role
  - store, forward frames
  - examine incoming frame's MAC address in each frame to decide where it should go
  - uses CSMA/CD when sending on a shared segment.
- *transparent*
  - hosts are unaware of presence of switches
- *plug-and-play, self-learning*
  - switches do not need to be configured

Switch forwarding table

Q: how does switch know PC3 reachable via interface 8, Pc2 reachable via interface 2?

- *A:* each switch has a switch table
- Each entry:
  - MAC address of host, interface to reach host, time stamp

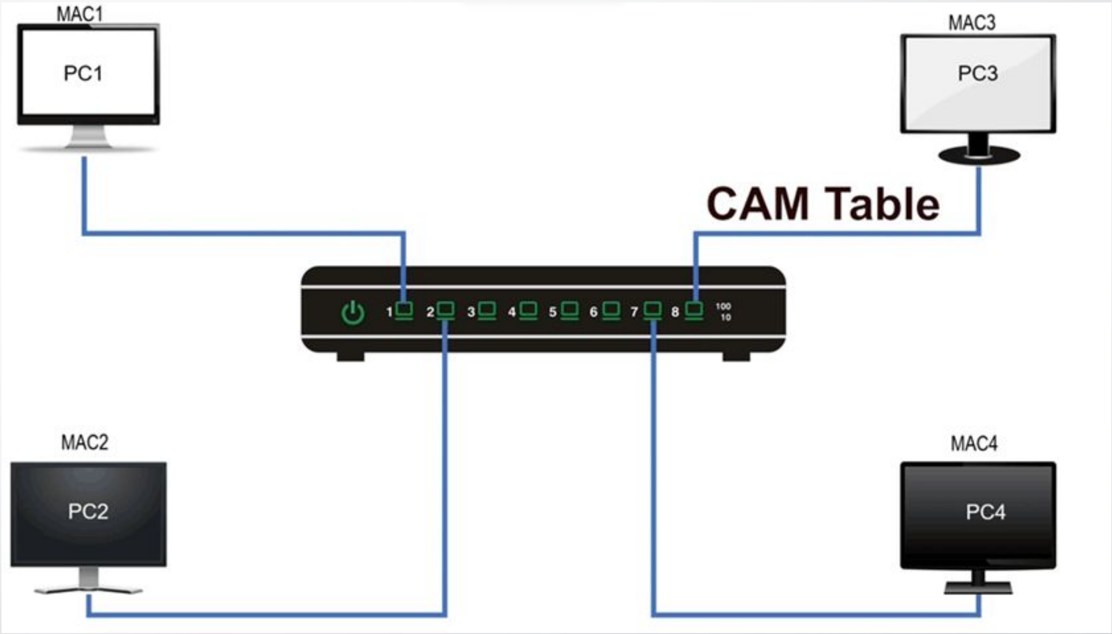


MAC address	Port	Aging
MAC1	Port1	24
MAC2	Port2	28
MAC3	Port3	29

Switch

- hosts have dedicated, direct connection to switch
- switches buffer packets

Q: how are entries created, maintained in switch table?

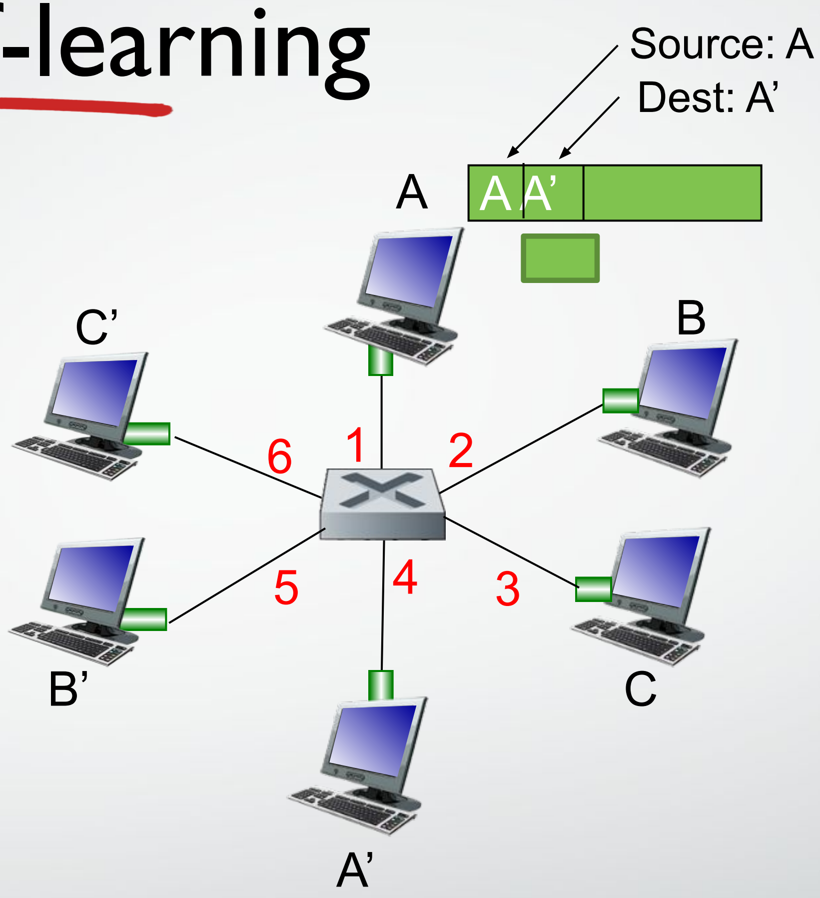


- Manual?
- Dynamic?

MAC address	Port	Aging
MAC1	Port1	24
MAC2	Port2	28
MAC3	Port3	29

Switch: self-learning

- The table is empty initially
- When it receives a frame, it looks at the **source MAC address**.
- It *learns* : "*This MAC address is reachable through this port.*"
    - Saves this info in its **switching table** for future use.



MAC addr	interface	TTL
A	*1*	*60*

Switch table (initially empty)

Switch: frame filtering/forwarding

when frame received at switch:

1. record incoming link, MAC address of sending host
2. index switch table using MAC destination address
3. if entry found for destination
   then {

   if destination on segment from which frame arrived
      then drop frame

      else forward frame on interface indicated by entry

   }

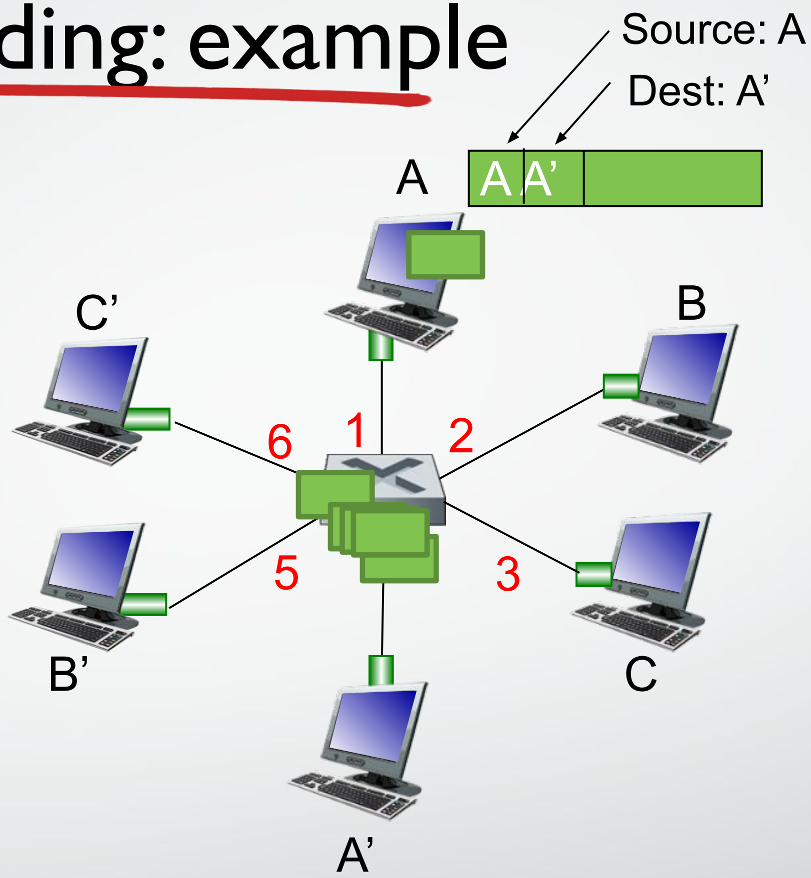
   else flood /* forward on all interfaces except arriving

   interface */

Self-learning, forwarding: example

- frame destination, A', location unknown:

  flood



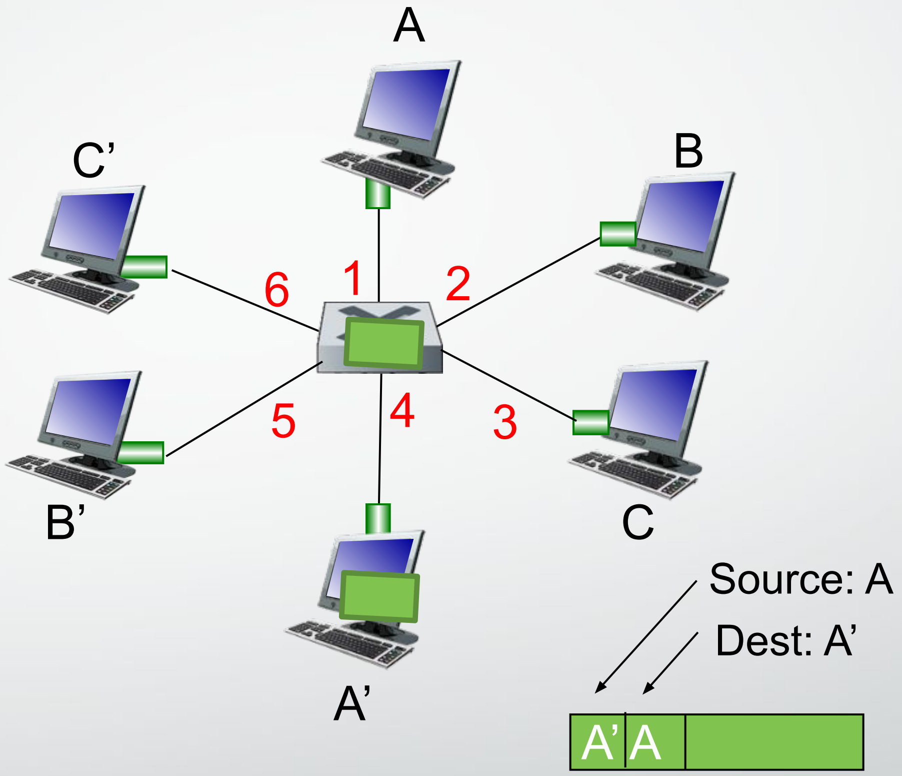
A' *4* *60*

MAC addr	interface	TTL
A	*1*	*60*

Self-learning, forwarding: example

- A' now wants to reply
- Switch looks at source MAC Address :
  learns
- Switch looks at destination MAC Address : ?
  - destination A location is known

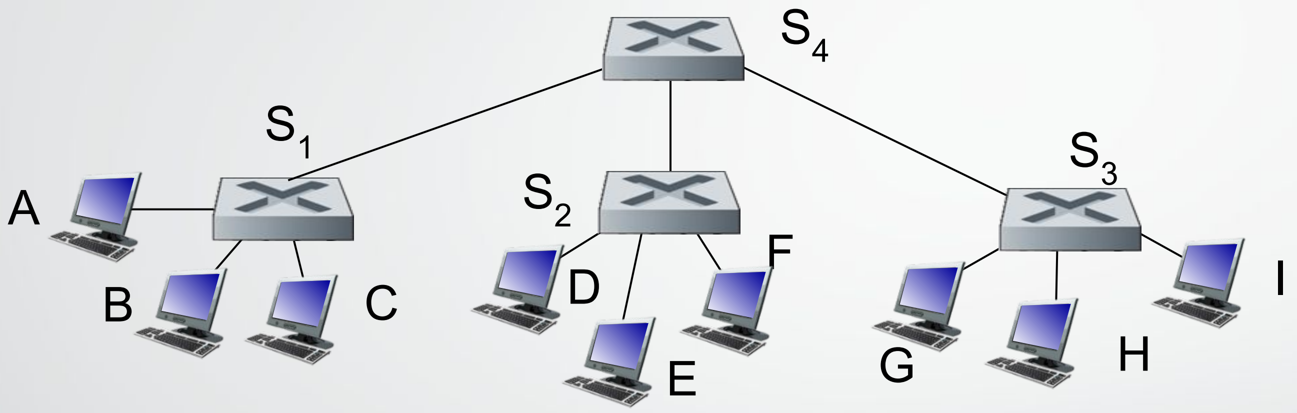
selectively sends/forwards



MAC addr	interface	TTL
A	*1*	*60*
A'	*4*	*60*

Interconnecting switches

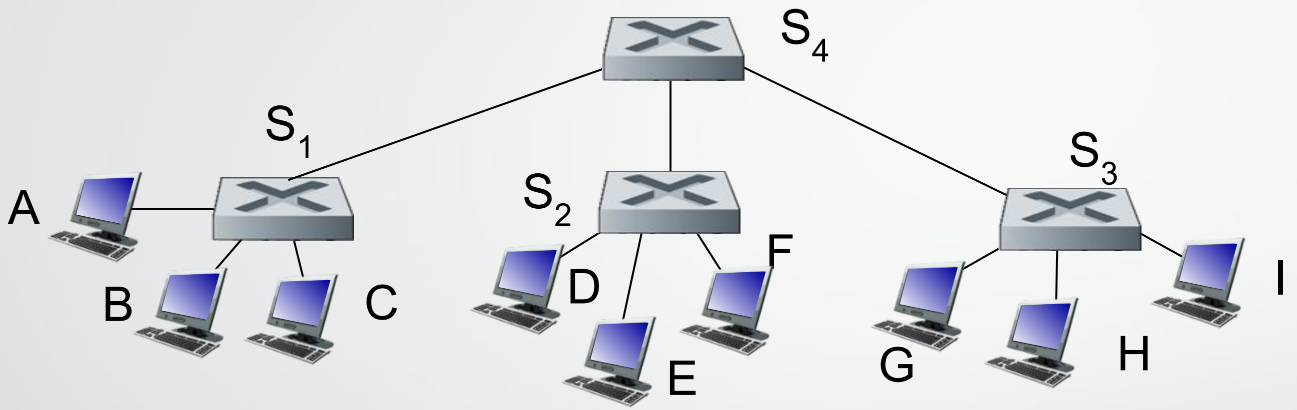
self-learning switches can be connected together:



Q: sending from A to G - how does S_1 know to forward frame destined to G via S_4 and S_3?

- *A:* self learning! (works exactly the same as in single-switch case!)

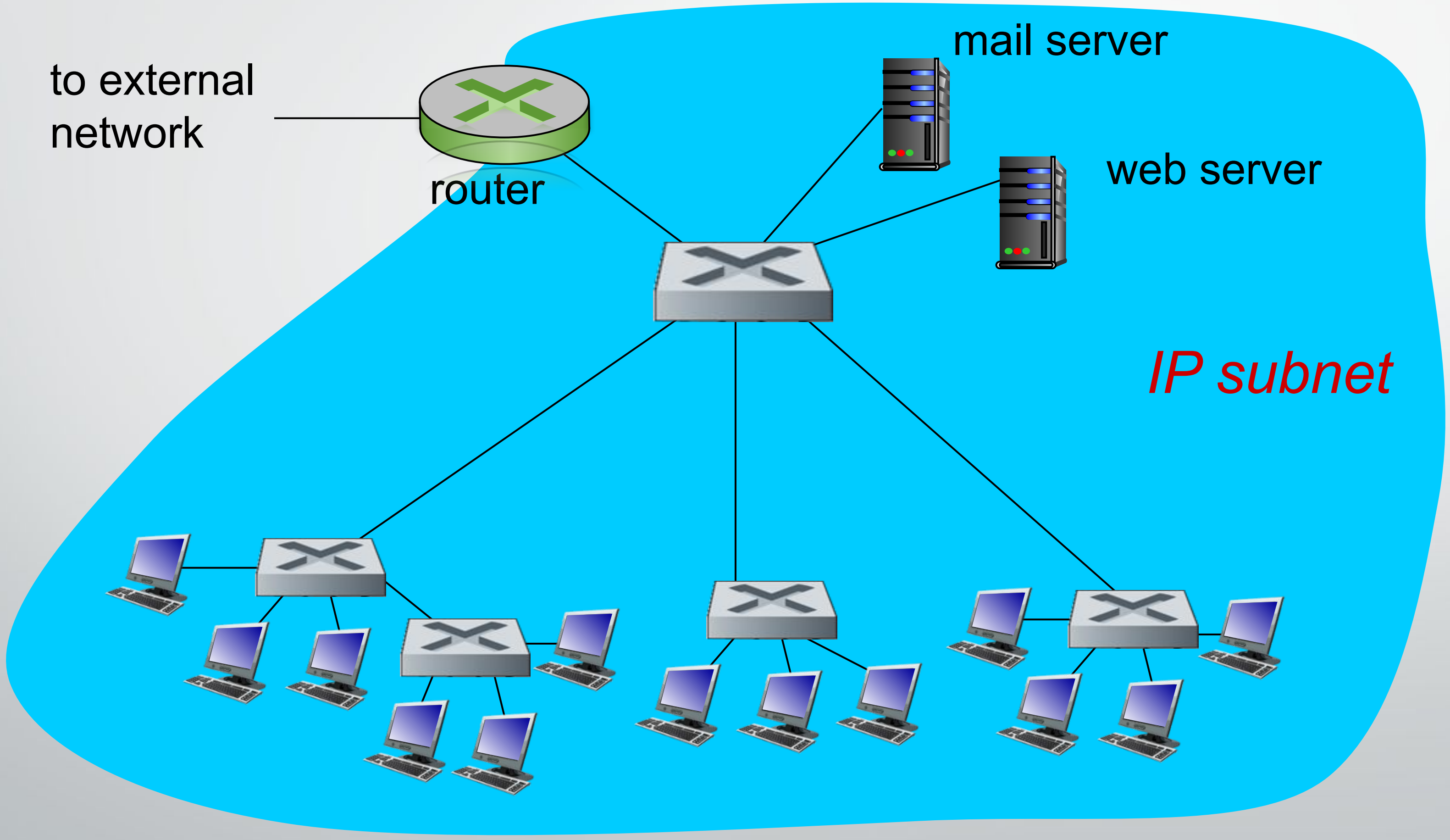
Self-learning multi-switch example



Suppose C sends frame to I, I responds to C

- Q: show switch tables and packet forwarding in S_1, S_2, S_3, S_4

Institutional network



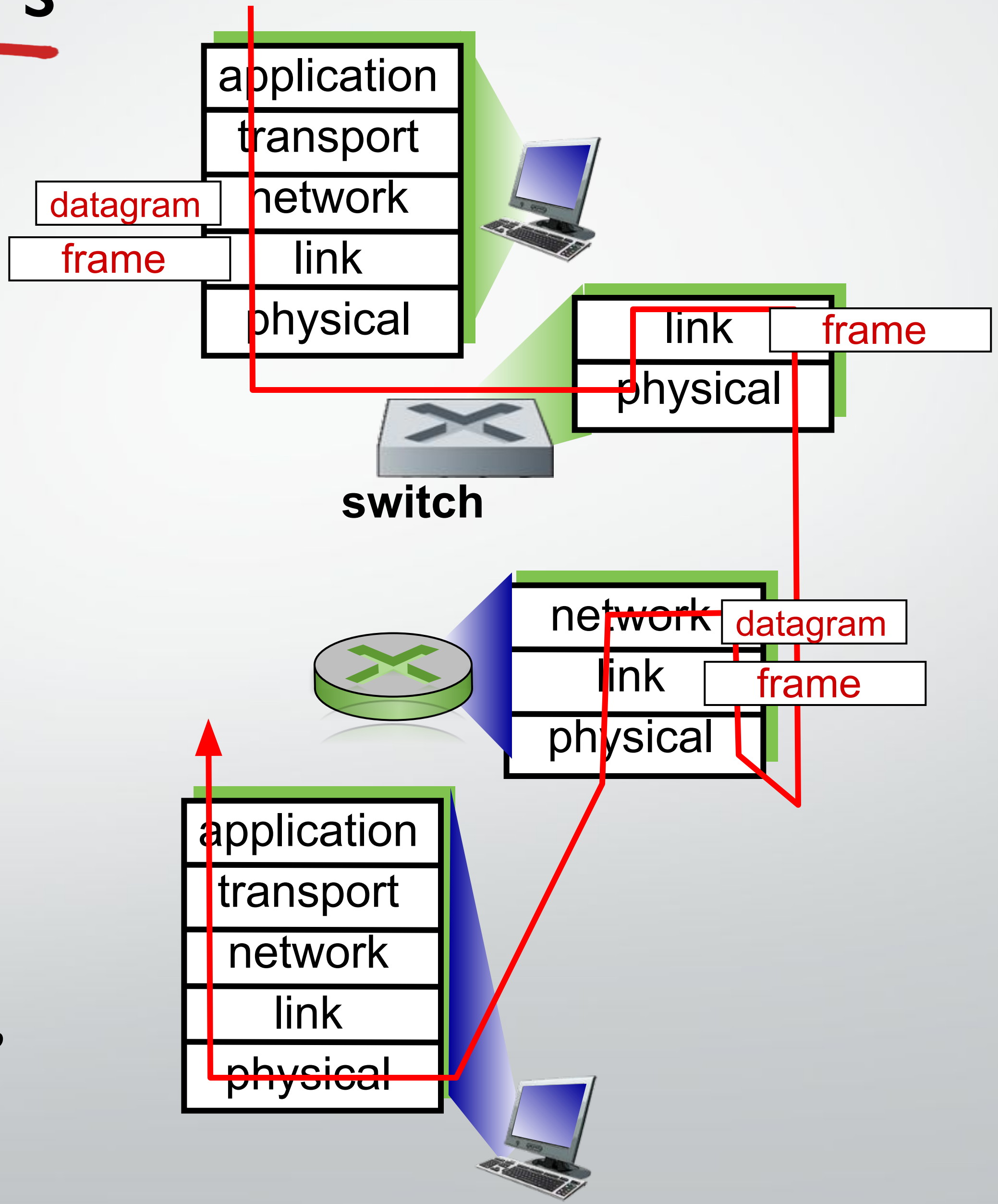
Switches vs. routers

both are store-and-forward:

- *routers:* network-layer devices (examine network-layer headers)
- *switches:* link-layer devices (examine link-layer headers)

both have forwarding tables:

- *routers:* compute tables using routing algorithms, IP addresses
- *switches:* learn forwarding table using flooding, learning, MAC addresses



THE END!